

a

F1a Pearson, three readings

held-out Pearson

ladder to $\lambda = 100$;
random at $\lambda = 1$ (and 0.1)
replaces random at 1e-3
stars per reading

log10 λ , none, mask, rand

b

F1b held-out loss

point loss

epoch 29 and
at min val loss

log10 λ

c

F1c edge recall

recall at degree

vs own target;
random arms also
vs the biological
graphs (from ckpt)

log10 λ

d

F1d divergence

divergence

floor at ~300:
does 10, 100
break it?

log10 λ

e

F1e gradient ratio

penalty / point

adds 10, 100
and random 1

probe epoch

f

F1f random control

Pearson, ep 29

biological vs random
vs none, AT $\lambda = 1$
(n = 3 each)

target

g

F2a Pearson, three readings

held-out Pearson

one column per mechanism;
stars vs none

mechanism, reach and direction (9 columns)

h

F2b held-out loss

point loss

epoch 29 and
at min val loss

mechanism

i

F2c edge recall

recall at degree

mask arms 1 by
construction on their
own support; score all
vs one-hop biological

mechanism

j

F2d val Pearson by epoch

Pearson

none, mask 1/2/3-hop,
KL 1 1/2/3-hop
(the curves move here
from F1 g)

epoch

k

F2e train Pearson by epoch

Pearson

same arms

epoch

l

F2f val loss by epoch

point loss

same arms

epoch

m

F3a 60 epochs, val Pearson

Pearson

none, mask, KL 1;
does KL 1 keep rising,
does none keep falling

epoch (0-59)

n

F3b placement curves

Pearson

val Pearson by epoch
for the placement arms

epoch

o

F3c discovery

recall of unseen edges

from best checkpoints:
regulatory head vs
TFLink-only edges;
null: degree-matched

top-k off-graph attention

p

F3d overruling

share with $|e| > 0.08$

from best checkpoints:
kept vs down-weighted
prior edges against
Costanzo digenic ϵ

$\alpha_{ij} / (1/d_{ij})$, binned

q

F3e divergence, all arms

divergence to biological $\bar{\lambda}$

none, mask, random,
symmetric: from ckpt,
fills the crosses of F1 d

arm

r

F3f seed table

per-run table of every
round-2 run, W&B ids,
as t2 today